# The complex $\ell_\infty(\Gamma)$ and $C(K)$ Mazur–Ulam conjecture was answered affirmatively

Literature-resolution packet for arXiv:1708.08538

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**Status: literature already answered (full conjecture).** Peralta explicitly proves the $\ell_\infty(\Gamma)$ clause in arXiv:1709.09584. Mori–Ozawa's Theorem 1 in arXiv:1804.10674 proves the Mazur–Ulam property for every unital complex $C^*$-algebra, which settles both $\ell_\infty(\Gamma)$ and $C(K)$.

## 1 Exact source conjecture

A Banach space $Z$ has the *Mazur–Ulam property* if, for every Banach space $Y$, each surjective isometry $S(Z) \to S(Y)$ extends uniquely to a surjective real-linear isometry $Z \to Y$. Jiménez-Vargas, Morales Campoy, Peralta, and Ramírez prove this for complex $c_0(\Gamma)$ in arXiv:1708.08538. On PDF page 15 they state:

> "We conjecture that the complex spaces $\ell_\infty(\Gamma)$ and $C(K)$ also satisfy the Mazur–Ulam property."

Thus the target is a two-clause affirmative conjecture, not merely the $\ell_\infty$ question highlighted by the later title.

## 2 Direct answer to the $\ell_\infty(\Gamma)$ clause

Peralta's arXiv:1709.09584 explicitly identifies arXiv:1708.08538 as the source of the conjecture. Its introduction says that the paper provides a proof, and Theorem 1.1 states that for every infinite set $\Gamma$ the complex space $\ell_\infty(\Gamma)$ has the Mazur–Ulam property. This is an exact match in the domain, quantifiers, target Banach space, and real-linear conclusion.

## 3 A theorem settling both clauses

Theorem 1 of Mori and Ozawa, arXiv:1804.10674, states that every unital complex $C^*$-algebra, regarded as a real Banach space, has the Mazur–Ulam property. Both spaces in the source conjecture fall under that theorem:

$$\ell_\infty(\Gamma) \cong C(\beta\Gamma) \quad \text{and} \quad C(K),$$

with their usual pointwise operations and supremum norm, are unital commutative complex $C^*$-algebras. Therefore Mori–Ozawa directly implies the Mazur–Ulam property for each. No additional separability, metrizability, or cardinality hypothesis occurs in their theorem.

| Clause | Answer | Provenance |
| --- | --- | --- |
| $\ell_\infty(\Gamma)$ | Peralta, Theorem 1.1; also Mori–Ozawa, Theorem 1 | Explicit source-aware answer, plus a later general implication |
| $C(K)$ | Mori–Ozawa, Theorem 1 | Direct $C^*$-algebra implication |

# 4 Scope and audit

The source conjecture as written has no unresolved clause. This packet does not claim that Tingley's problem is solved for arbitrary Banach spaces; only the two named domain classes are closed here.

The official arXiv PDFs of 1708.08538, 1709.09584, and 1804.10674 were inspected. The source conjecture occurs on PDF page 15; the explicit $\ell_\infty$ answer occurs on PDF page 2 of 1709.09584; and the unital $C^*$-algebra theorem occurs on PDF page 1 of 1804.10674. The first match is explicitly source-aware. The second is recorded as a theorem implication; no claim is made here that Mori–Ozawa intended to answer this exact source conjecture. Human review should confirm only the bibliographic linkage and this provenance distinction.

# References


[1] A. Jiménez-Vargas, A. Morales Campoy, A. M. Peralta, and M. I. Ramírez, *The Mazur–Ulam property for the space of complex null sequences*, arXiv:1708.08538; Linear Multilinear Algebra 67 (2019).

[2] A. M. Peralta, *Extending surjective isometries defined on the unit sphere of* $\ell_\infty(\Gamma)$, arXiv:1709.09584; Rev. Mat. Complut. 32 (2019), 99–114.

[3] M. Mori and N. Ozawa, *Mankiewicz's theorem and the Mazur–Ulam property for* $C^*$*-algebras*, arXiv:1804.10674; Studia Math. 250 (2020), 265–281.